



TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2016
Higher 2

CANDIDATE
NAME

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CENTRE
NUMBER

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INDEX
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BIOLOGY

9648/02

Core Paper

Wednesday, 31 August 2016

Paper 2

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show any working or if you do not use appropriate units.

At the end of the examination, fasten the writing papers securely and hand up separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Q1	/ 10
Q2	/ 9
Q3	/ 10
Q4	/ 10
Q5	/ 11
Q6	/ 9
Q7	/ 12
Q8	/ 9
Essay	/ 20
Total	/ 100

1 Fig. 1.1 shows the cell cycle.

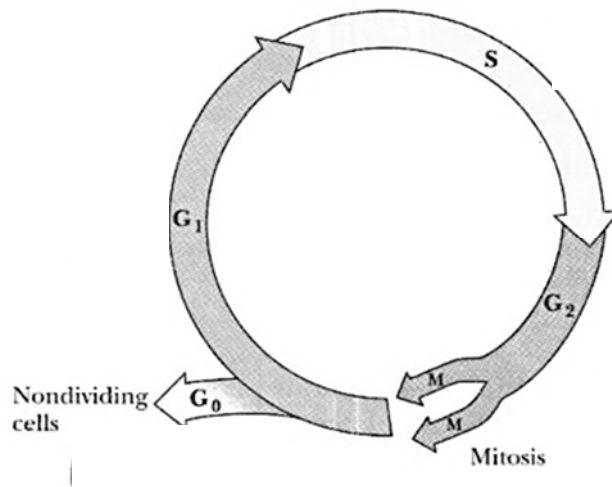


Fig. 1.1

(a) Describe the events that occur during **S** and **G₂** phases. [2]

1. **S (synthesis) phase:** Replication of DNA occurs. This doubles the DNA content of the cell
2. **G₂ (second "gap") phase:** Synthesis of proteins – These proteins include histones, ribosomal proteins and proteins which later make up the microtubules of the spindle.
OR
Formation of new organelles (e.g. mitochondria, centrioles, chloroplasts, ribosomes).

(b) Mitosis, a form of nuclear division, occurs after the **G₂** phase. This process produces genetically identical cells.

Explain the need for the production of genetically identical cells. [2]

1. Genetically identical cells contain the same number and type of chromosomes.

Mitosis is needed to produce genetically identical cells

2. during the growth and development of a multicellular organism, e.g. in the development of a fertilized egg into an adult human being.
3. And during the replacement of worn-out parts of the body.
4. Mitosis is the basis of asexual reproduction, where parents and offspring are genetically identical.

Sometimes, errors during cell division can lead to genetically non-identical cells, causing the production of polyploid cells. Polyploidy is common in plants, and one such example is the tetraploid (4n) snapdragons.

(c) State what is meant by *polyploidy*. [1]

Cells with one or more extra sets of chromosomes [3n, 4n etc.].

(d) Explain how a tetraploid cell may be formed due to errors during mitosis. [4]

1. DNA replication occurs prior to mitosis,
2. hence the DNA content is doubled in the parental cell.
3. However, during metaphase
4. spindle fibers fail to attach to chromatids / spindle fibers are not formed / spindle fibers are not stretched across the cell properly
5. Thus, during anaphase, centromeres divide,
6. But sister chromatids are not pulled to opposite poles
7. Cytokinesis fails to occur.
8. Thus, there is double the number of chromosomes in the daughter cell.

Triploid plants, such as bananas, are sterile as they do not produce gametes, and are hence seedless.

(e) Suggest how farmers can grow bananas without seeds. [1]

Stem cuttings / plant cloning / micropropagation / plant tissue culture

[Total: 10]

2 Fig. 2.1 is a diagram showing the process of translation in a cell.

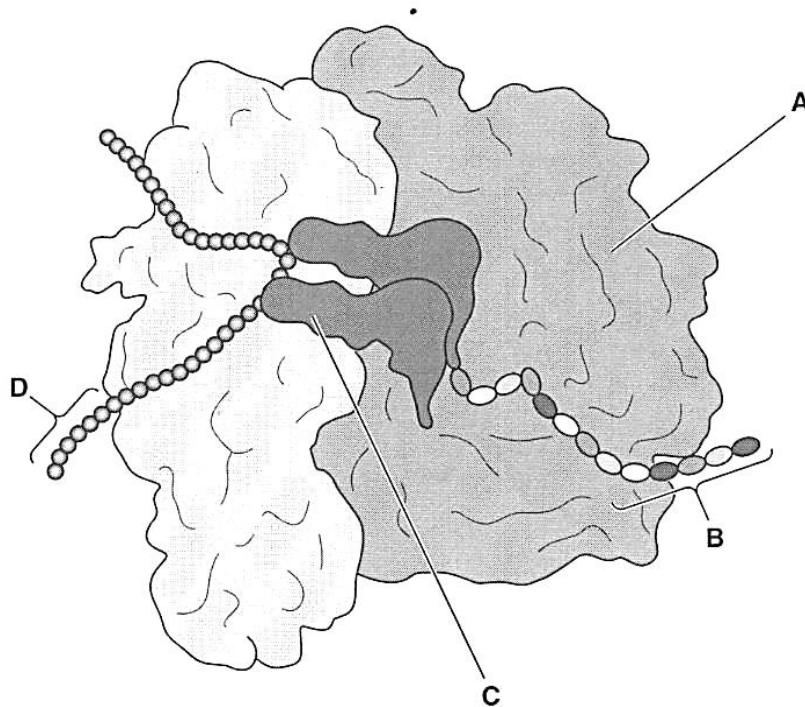


Fig. 2.1

(a) Identify **A**, **B** and **C**. [3]

A – large (60S) subunit of ribosome

B – polypeptide chain / chain of amino acids / amino acid residues

C – tRNA

(b) Briefly describe how the amino acids are joined in the correct order during translation. [4]

1. The first amino acyl-tRNA complex with the anticodon UAC binds to the start codon (AUG) on mRNA. This complex is positioned at the “P” site of the ribosome.
2. tRNA with the anticodon UAC always carries the amino acid methionine. Methionine is always the first amino acid in a polypeptide chain.
3. The second amino acyl-tRNA complex binds to mRNA. Its anticodon is complementary to the second codon on mRNA. The complex is held at the “A” site of the ribosome.
4. Formation of a peptide bond between the first and the second amino acids then occurs, utilising energy from hydrolysis of GTP (a molecule similar to ATP). This is catalysed by the peptidyl-transferase (RNA enzyme) which is present on the large subunit of the ribosome.
5. Once the peptide bond is formed, the ribosome moves along the mRNA to the next codon. / 5' to 3' direction / that the ribosome “reads” the mRNA in the 5' to 3' direction.
6. The first tRNA, now at the “E” site, is then released into the cytoplasm for it to be recycled. It can be attached again to their respective amino acids.

7. The second amino acyl-tRNA complex thus moves from the “A” site to the “P” site, leaving the “A” site empty for the next amino acyl-tRNA complex, with anticodon complementary to the third codon along the mRNA.
8. (Idea : each tRNA carries a specific amino acid)
Note that polypeptide chain is synthesized from the amino to carboxyl end. The process is repeated until the ribosome reaches the “stop” codon on the mRNA.

Parasitic infections affect about one billion people. Scientists are exploring new anti-parasitic drugs that target the translation process in parasites. One such target is the aminoacyl-tRNA synthetases.

(c) Explain how drugs targeting aminoacyl-tRNA synthetases are effective. [2]

1. Drug inhibits aminoacyl-tRNA synthetases. (OWTTE)
2. Aminoacyl-tRNA synthetases cannot ensure the correct / specific amino acid is joined to a specific tRNA. / Therefore cannot form the specific aminoacyl tRNA complexes.
3. It is unable to catalyse formation of ester / covalent bond between tRNA and amino acid with expenditure of ATP.
4. Therefore cannot form functional protein in the parasite.

(d) Suggest **one** way how the drugs act on aminoacyl-tRNA synthetases. [1]

1. Bind to anticodon binding site to block binding to tRNA
2. Bind to the amino acid binding site to block binding of amino acid
3. Bind to an allosteric site to change shape of active site.
4. Bind to ATP binding site to block the binding to ATP

3 (a) Briefly describe how specific transcription factors function to affect the rate of transcription. [3]

1. The binding of activators to enhancers
2. make it easier for RNA polymerase II and general transcription factors to bind to DNA,
3. speed up assembly of transcription initiation complex
4. thus increasing the rate of transcription
5. Repressors bind to silencers

6. to slow down/stop transcription.

Fig. 3.1 illustrates the processing of an RNA transcript that must take place for translation to occur.

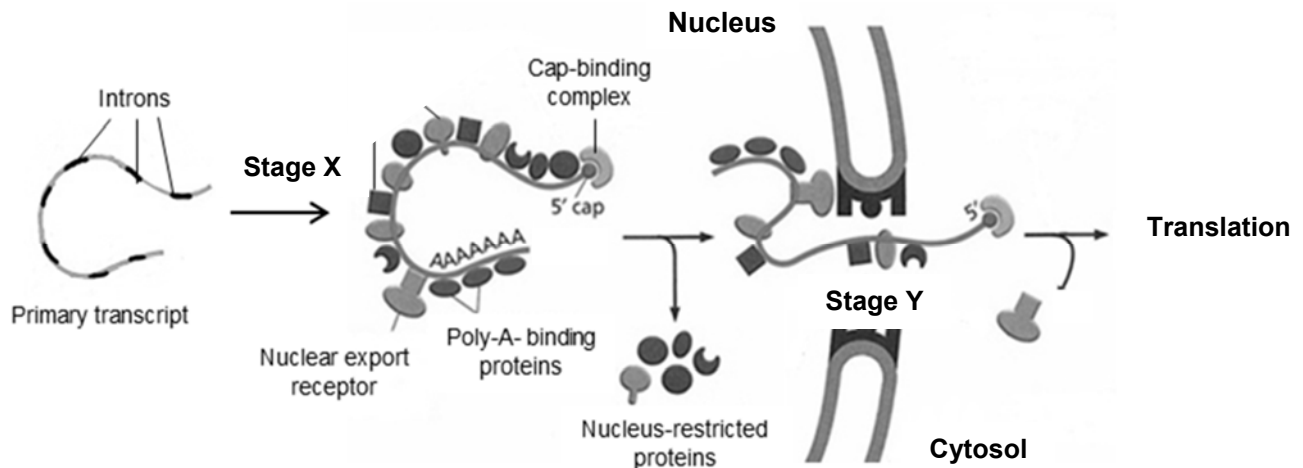


Fig. 3.1

With reference to Fig. 3.1,

(b) outline the process in **Stage X** that results in a continuous mRNA coding sequence to be formed; [3]

1. RNA splicing
2. Small nuclear RNA (snRNA) base pair with the splice sites
3. Allowing small nuclear ribonucleoproteins (snRNPs) to recognise and bind the splice sites
4. Several snRNPs join with additional proteins to form an even larger assembly called a spliceosome
5. The spliceosome interacts with the splice sites at the ends of an intron and cuts at specific points to release the intron
6. then immediately joins together the two exons that flanked the intron

(a) suggest the roles of **two** named proteins and/or protein complexes; [2]

(any 2)

1. Nuclear export receptor – facilitates export / regulates exit of mature mRNA from nucleus to cytosol
2. Nucleus-restricted proteins – stabilises/protects the mRNA in the nucleus
3. 5'cap binding complex – (related to functions of 5'cap) helps protect mRNA from degradation/ serves as part of a signal for attachment of ribosomes for translation in the cytoplasm
4. Poly-A-binding proteins – (related to functions of poly A tail) helps prevent degradation of the mRNA and helps ribosomes attach to it/

helps to facilitate the export of mRNA from the nucleus to cytosol

(b) explain **two** processes that will **not** take place in a bacterial cell. [2]

(any 2)

1. RNA splicing– spliceosome/ introns are absent in bacteria chromosome
(Reject: **enzymes/ mechanism are not present**)
2. Addition 5'cap/ poly-A tail – bacteria chromosomes are circular/ not linear
3. Involvement of nuclear export receptor/ nucleus-restricted proteins – bacteria do not have nucleus

[Total: 10]

- 4 A pure-breeding white squash fruit was crossed with a pure-breeding green squash fruit, and the resulting F1 generation was all white. In the F2 generation, the following squashes and the corresponding numbers were produced.

White squash 603
Yellow squash 152
Green squash 49

The control of squash fruit color is an example of epistasis.

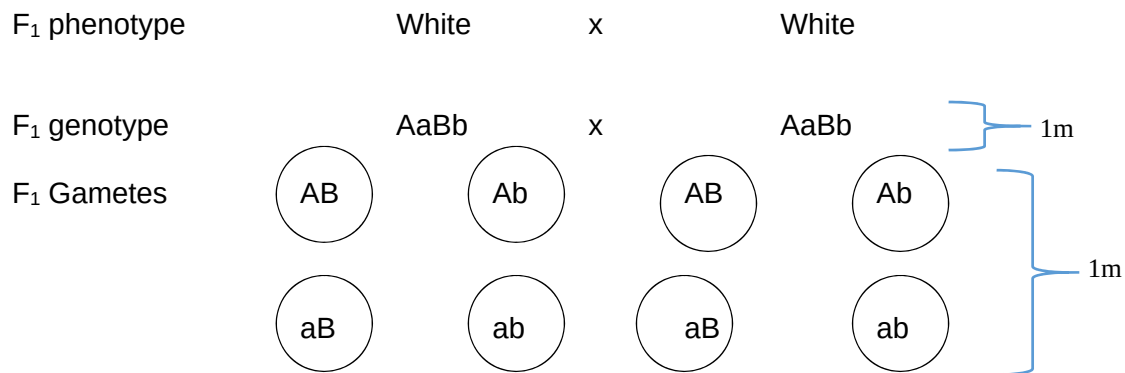
(a) Using the symbols **A**, **a**, **B** and **b**,

(i) state and explain the specific type of epistasis in the control of squash fruit color.

1. Dominant epistasis. [1]
2. The presence of one copy of the epistatic allele (A) inhibits the phenotypic expression of the hypostatic gene (B/b). [1/2]
3. Squashes that possess 1 copy of the epistatic allele (A) will be white colour even if they carry the alleles B or b for coloured pigment / squash / green or yellow squash. [1/2]
4. The gene product of allele A inhibits the phenotypic expression of the gene for coloured pigment / squash / green or yellow. [1/2]
5. Squashes with the dominant allele A will be white. Squash with recessive alleles aa and dominant allele B will be yellow and squash with recessive alleles aa and recessive alleles bb will be green. For these squash, there is no inhibition on the phenotypic expression of gene B/b. [1/2]

(ii) draw a genetic diagram to explain the results of the F₂ generations. [4]

Let A represent the dominant allele resulting in white squash.
 a represent the recessive allele resulting in colored squash.
 B represent the dominant allele resulting in yellow squash
 b represent the recessive allele resulting in green squash



Punnett square to show fusion of gametes [2m]

				
	 White	 White	 White	 White
	 White	 White	 White	 White
	 White	 White	 Yellow	 Yellow
	 White	 White	 Yellow	 Green

F₂ phenotypes White : Yellow : Green

F₂ phenotypic ratio 12 : 3 : 1

Observed nos 603 : 152 : 49

The control of squash fruit color is an example of discontinuous variation, while seed color in wheat is an example of continuous variation.

(b) State **two** differences between discontinuous variation and continuous variation. [2]

DISCONTINUOUS VARIATION	CONTINUOUS VARIATION
1. <u>Discrete phenotypic classes</u> with no intermediates are observed.	A <u>range of phenotypes</u> is observed.
2. Characteristic is usually controlled <u>by one or two major genes</u> , which may have two or more allelic forms.	Characteristic is usually controlled by <u>a large number of genes</u> .
3. Effect of <u>individual genes</u> can be observed.	Effect of individual genes <u>cannot be observed</u> .
4. Effect of genes is <u>not</u> additive.	Effect of genes is <u>additive</u> .
5. The environment has a <u>small effect</u> on the <u>phenotype</u> . (reject: no effect)	<u>Environment</u> has a <u>large effect</u> on the <u>phenotype</u> .

[Total: 10]

[Turn over

- 5 The unicellular green alga, *Chlorella*, was originally studied for its potential as a food source. Although large-scale production proved to be uneconomic, the many health benefits provided by *Chlorella* mean that it is now mass produced and harvested for use as a health food supplement.

In one study into the productivity of *Chlorella*, carbon dioxide concentration was altered to investigate its effects on the light-independent stage of photosynthesis.

- A cell suspension of *Chlorella* was illuminated using a bench lamp.
- The suspension was supplied with carbon dioxide at a concentration of 1% for 200 seconds.
- The concentration of carbon dioxide was then reduced to 0.03% for a further 200 seconds.
- The concentrations of RuBP and glycerate-3-phosphate (GP) were measured at regular intervals.
- Throughout the investigation the temperature of the suspension was maintained at 25 °C.

The results are shown in Fig. 5.1.

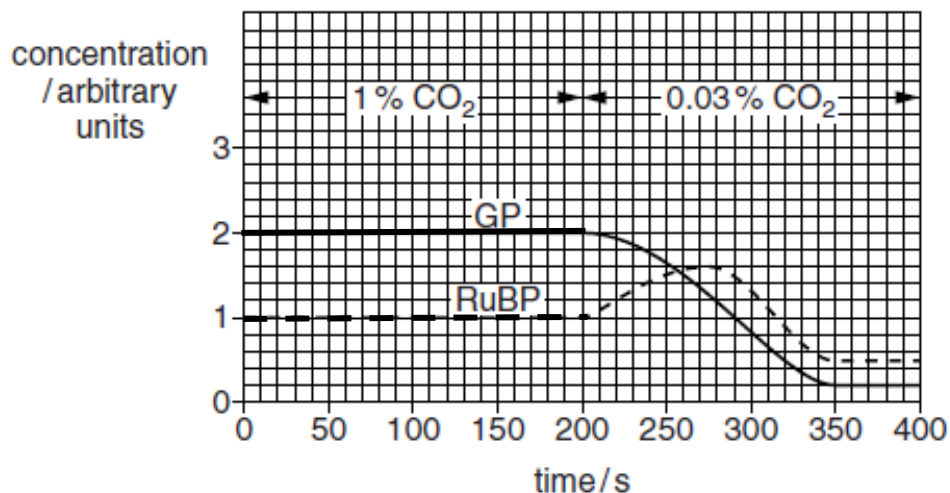


Fig. 5.1

(a) **State precisely** where RuBP and GP are located. [1]

- **Stroma of chloroplast**

(b) **Explain** why the concentration of RuBP changed **between 200 and 275 seconds**. [2]

State the change in conc of RUBP (QF)

1. **RuBP increased from 1 to 1.6 AU**

Explain

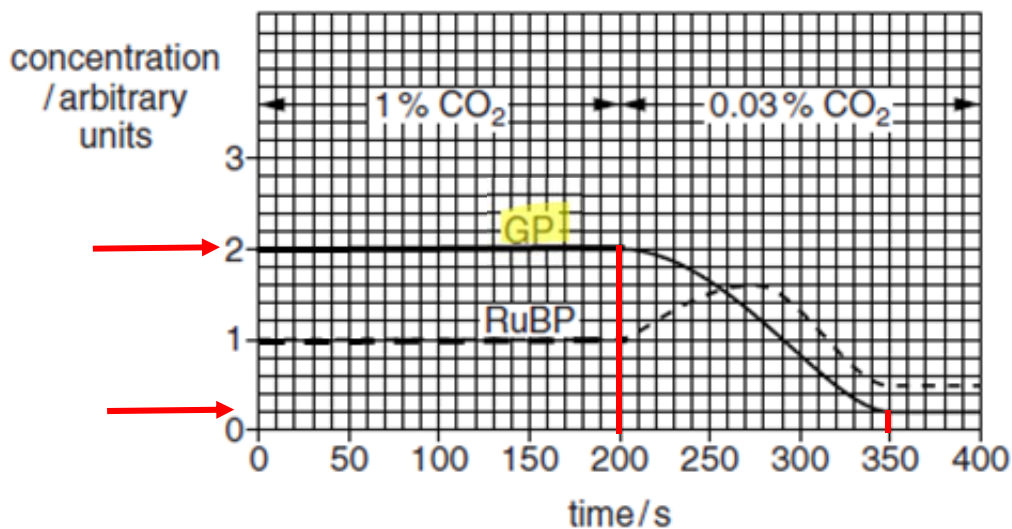
2. **Less needed for carbon dioxide fixation**
3. **Due to lowered carbon dioxide concentration at 0.03%**
4. **While RuBP continues to be regenerated from glyceraldehyde- 3- phosphate**

- (c) Calculate the **rate of decrease per second** in the concentration of GP **between 200 and 350 seconds**.

Show your working and present your answer to **two decimal places**. [1]

$$\frac{2.0 - 0.2}{350 - 200}$$

Answer: 0.01 arbitrary units per second



- (d) Explain how the **decrease** in the **concentration** of **GP** leads to a **decreased harvest** for commercial suppliers of *Chlorella*. [2]

1. There will be reduced glyceraldehyde-3-phosphate
 2. Decreased conversion to glucose/carbohydrates/starch
 3. Leading to reduced growth
- Thus reduced dry mass of Chlorella

- (e) Production of GP **in plant cells** leads to two types of carbohydrates, one for storage and the other for structural support. Describe the **differences** in structure **between** these **two carbohydrates**. [3]

State the two carbohydrates:

1. Starch and cellulose are formed
(2 differences:)

Describe two differences

2. Monomer of starch is α -glucose, while for cellulose it is β -glucose
3. α (1,4) glycosidic bonds are formed between the monomers in starch, while β (1,4) glycosidic bonds are formed in cellulose

4. Hydrogen bonds forming cross-linkage between neighbouring β -glucose chains are present in cellulose but absent in starch
5. Branched structure of amylopectin is found in starch, while only straight/linear/no branching occurs in cellulose
6. Hydroxyl groups project inside the helix in amylose of starch, while hydroxyl groups project outwards in each chain (thereby allowing cross-linking between chains) in cellulose
7. The β -glucose chains are associated into groups to form microfibrils, which then combine to form macrofibrils in cellulose, but this is not observed in starch

GP is also formed during cellular respiration.

(f) **Outline** the fate of GP during respiration. [2]

1. During glycolysis,
2. it is converted to pyruvate
- 3a. Forming ATP
- 3b. via substrate level phosphorylation
This occurs in cytosol/ cytoplasm

[Total: 11]

6 Processes that contribute to genetic variation are of significance to the survival of organisms. Some of these processes result in new combination of alleles in offspring.

(a) **Describe** the ways in which new combinations of alleles can arise during gamete formation. [3]

1. Crossing over between homologous chromosomes during Prophase I.
2. There is exchange of corresponding sections / corresponding alleles
3. Random / Independent arrangement of homologous chromosomes during Metaphase I
4. Random / Independent separation of homologous chromosomes during Anaphase I.
5. random/ independent arrangement of chromosomes at the equator of the spindle during Metaphase II,
6. and subsequent separation of chromatids of these chromosomes during Anaphase II.

[Turn over

Other processes, like mutation, can also contribute to genetic variation. However, most mutation events are harmful to the organism and accumulation of these mutations may lead to uncontrolled cell division, resulting in cancer.

(b) **Explain** how uncontrolled cell division can result in cancer. [3]

1. Cancer is the uncontrolled growth of cell that results from the failure of control of cell division / uncontrolled cell division.
2. and the inability of cells to undergo apoptosis (programmed cell death).
3. This forms malignant tumours when the cells:
4. Stimulate the growth of blood vessels towards themselves (angiogenesis) to obtain the necessary nutrients they need to thrive.
5. They undergo invasion: occurs when cancer cells directly migrate to and penetrate into neighbouring tissues
6. And metastasis, which occurs when cancer cells from a tumour penetrates into the blood or lymphatic system, travels through the bloodstream and invade normal tissues elsewhere in the body.

In cancer patients, one of the genes that is frequently mutated is the *K-ras* gene, that codes for the K-ras protein.

In a normal cell, the K-ras protein is activated when an epidermal growth factor (EGF) binds to its receptor (EGFR) as shown in Fig. 6.1A. However, when an EGFR inhibitor, EGFR-I binds to EGFR, the cell signaling pathway is inhibited as shown in Fig. 6.1B.

When the K-ras gene is mutated, the mutated K-ras protein continues to result in signal transduction, even in the presence of the inhibitor EGFR-I, as shown in Fig. 6.1C.

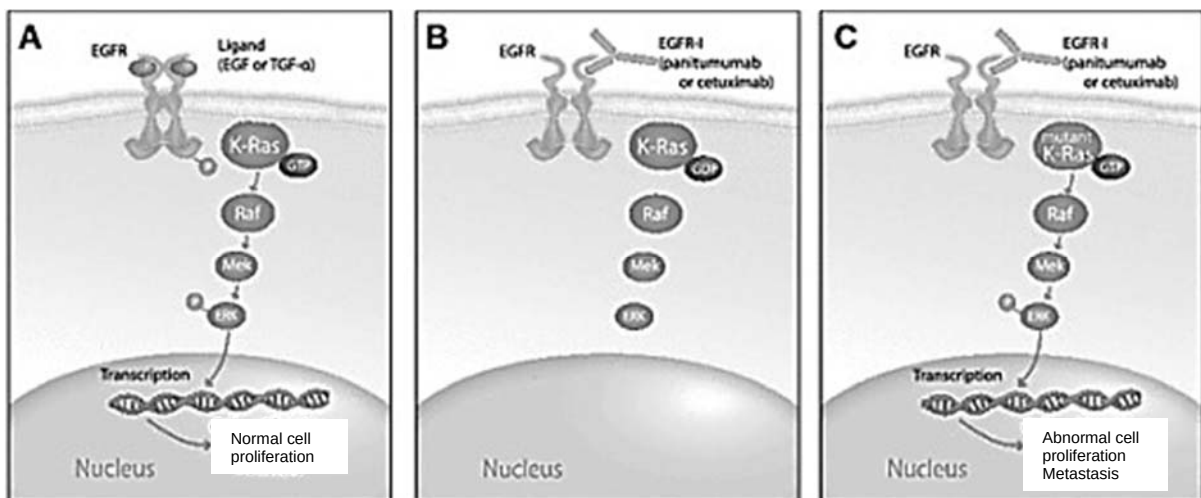


Fig. 6.1

- (c) With reference to Fig. 6.1, **explain** whether the *K-ras* gene undergoes a gain-of-function or loss-of-function mutation to result in uncontrolled cell division. [3]

1. Gain-of-function mutation.

2. Gain-of-function mutation in a single copy of the *K-ras* gene

3. results in a mutant *K-ras* protein (mutant protein must mention at least once).

4. The mutated *K-ras* protein is activated even in the absence of epidermal growth factor / in the presence of the EGFR inhibitor.

5. It binds to GTP to activate the signal transduction pathway to result in abnormal / excessive / uncontrolled cell proliferation, even in the absence of growth factors.

[Total: 9]

- 7 Cytochrome c is found in the mitochondria of every aerobic eukaryote — animal, plant, and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related. 1

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. These molecules are homologous as an indication that all organisms share a common ancestor. 5

Table 7.1 shows the first 18 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath. 10

A ditto (“) indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochromes have several amino acids that precede the sequence shown here (indicated by <<<).

Table 7.1

Cytochrome c	Amino acid																
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln
Pig		“	“	“	“	“	“	“	“	“	“	Val	Gln	“	“	Ala	“
Chicken		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“
Turkey		“	“	Ile	“	“	“	“	“	“	“	Val	Gln	“	“	“	“
Dogfish		“	“	“	“	“	“	“	“	Val	“	Val	Gln	“	“	Ala	“
<i>Drosophila</i>	<<<	“	“	“	“	“	“	“	“	Leu	“	Val	Gln	Arg	“	Ala	“
Wheat	<<<	“	Asn	Pro	Asp	Ala	“	Ala	“	“	“	Lys	Thr	“	“	Ala	“
Yeast	<<<	“	Ser	Ala	Lys	“	“	Ala	Thr	Leu	“	Lys	Thr	Arg	“	“	“

[Turn over

(a) With reference to Table 7.1, **explain** what is meant by “*These molecules are homologous*” (line 7). [2]

1. *[Meaning of homology in evolution]*
The cytochrome molecule of the various organisms is a shared characteristic / common molecule descended / derived from a common ancestor;
2. *[GS from Table 7.1]*
The cytochrome molecules share the same amino acids at many positions (not all positions) / are conserved;
3. *[QF from table 7.1]*
For all organisms, the molecules have the same amino acid at positions 1, 6, 10, 14, 16 from the N-terminal

[MUST QF ALL] / ALSO ACCEPT THE QF OF ALL 5 AMINO ACID]

(b) **State one limitation** of only using difference in amino acid sequences to establish phylogenetic relationship among the organisms. [1]

1. Protein represents only a very small percentage of the genetic information in the organisms

OR

2. Is not representative of the entire genome / only coding sequences are compared

OR

3. Data from protein may not be accurate / cause errors in the order of branching

[Reason : the method in determining the amino acids is not highly accurate]

Cytochrome c is an ancient molecule which has coding and non-coding regions. It evolved very slowly. Even after more than 2 billion years, one-third of its amino acid unchanged. The conserved sequences are a great help in working out the evolutionary relationships between distantly-related organisms like fish and humans.

(c) **Suggest one reason** why the apparent rate of accumulation of gene mutations in coding regions is slower than in non-coding regions. [1]

Any one:

1. Mutations in coding regions may result in non-functional protein
2. May cause organism to die and mutations not passed to next generation / offspring.

OR

3. Mutations in coding regions are more likely to be subjected to natural selection.

4. If the mutation places the organism at selective disadvantage, it is less likely to be passed on. Hence, slower accumulation of mutations.

OR

5. Mutations in coding regions will be expressed in the phenotype and will be subjected to natural selection
 6. if it confer selective disadvantage, it will not survive and the mutation is not passed on.

OR

7. While mutations at non-coding regions tend to be neutral / does not place organism at selective disadvantage.
 8. The mutations are passed down to offspring as the individual is not affected. Hence faster accumulation of mutations.

Fig. 7.1 shows how the comparison of cytochrome c sequences from different species has been used to order the divergence of species in relative time. The sequence of the cytochrome c gene was compared between each pair of organism.

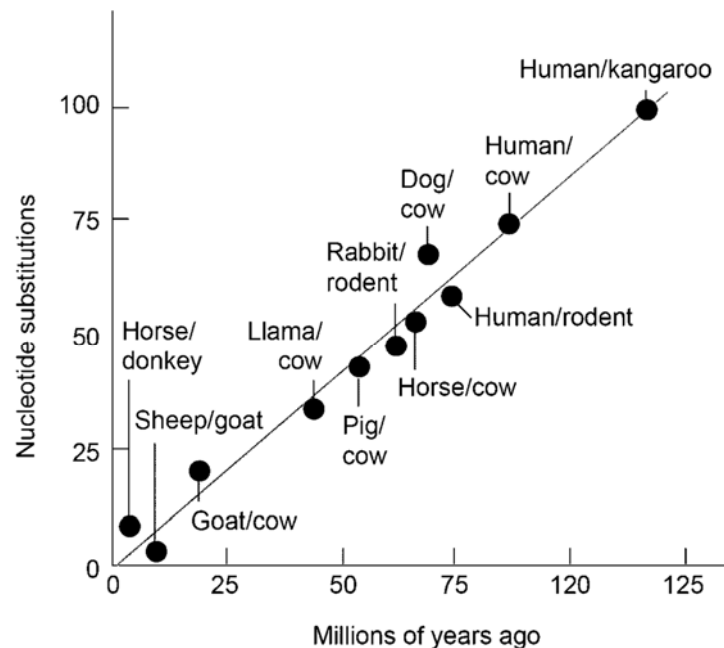


Fig. 7.1

(d) **Explain** how the information in Fig. 7.1 supports the Neutral theory. [3]

G.S-Q.F:

1. As the time increases from 0my to 123my, the number of substitutions in cytochrome c increases from 0a.u. to 100a.u.

Explain what the trend means:

2. The trend gives rise to a plotted straight line.
 3. This means that the rate of mutation is steady / constant.

4. Because mutations are silent and will be passed on to the following generation.
5. So it supports the neutral theory of molecular evolution which claims that genetic changes in genes do NOT confer selective advantage or disadvantage and there is no effect on the phenotype.

(e) **Describe how** the **cytochrome c molecular clock** can be used to **estimate how long** ago the **cow** and **kangaroo** may have **diverged** from each other. [2]

1. Compare cytochrome c gene and obtain the nucleotide difference / total number of nucleotide substitutions in the cytochrome c gene between the cow and kangaroo.
2. Obtain the corresponding value of time from the X-axis based on the number of nucleotide differences (from the Y-axis)

The *cytochrome c* gene shows that all organisms evolved from a common ancestor.

(f) **Explain** why a **population** is the **smallest** unit that can **evolve**. [3]

1. Microevolution is a change in allele frequencies over generations
2. And only the allele frequencies of a population can change.
3. This is due to natural selection selecting individuals in the population
4. based on their characteristics which affect their chances of survival and reproductive success. / natural selection selecting individuals in the population which are of selective advantage.
5. This, together with gene flow and genetic drift, changes allele frequencies in populations.
6. Whereas an individual's allele frequencies cannot change throughout the lifetime of the individual.

[Total: 12]

- 8 Acetylcholine is a neurotransmitter that binds to post-synaptic membrane and stimulates the production of nerve impulses.

Gamma-aminobutyric acid (GABA) is another neurotransmitter that is produced by certain neurones in the brain and spinal cord. It binds to post-synaptic membrane and inhibits the production of nerve impulses on the post-synaptic neurone.

Fig. 8.1 shows a synapse involving three neurones

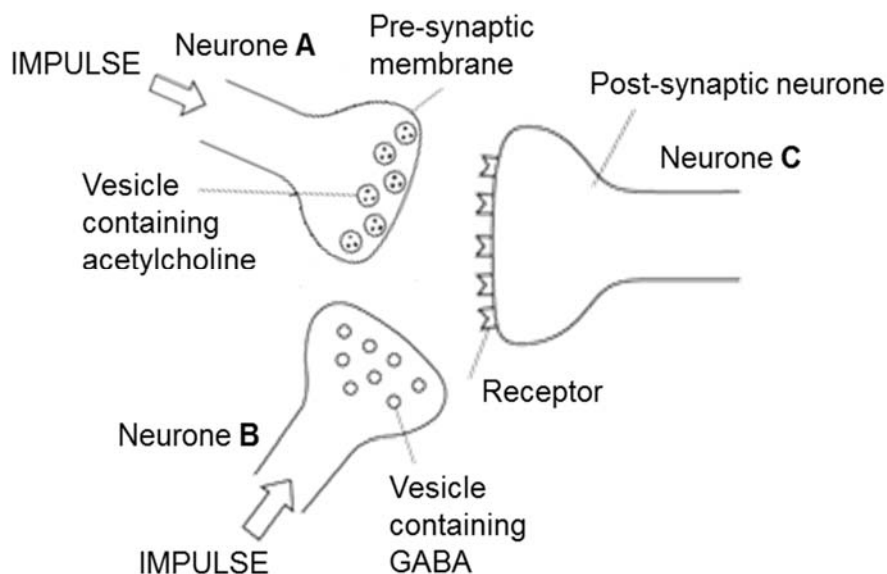


Fig. 8.1

- (a) Depolarization of pre-synaptic membrane of neurone A causes the opening of voltage-gated calcium channels. **Describe** the **role of calcium ions** in **synaptic transmission**.
[8]

1. Opening of voltage-gated Ca^{2+} channels cause the Ca^{2+} ions to diffuse rapidly into the knob.
2. This causes the synaptic vesicles containing acetylcholine to fuse with the presynaptic membrane,
3. releasing acetylcholine into the synaptic cleft by exocytosis.
4. Acetylcholine binds to receptors on post-synaptic membrane

(b) **State** and **describe** the function of **one** type of protein molecule on the post-synaptic membrane. [2]

1. Receptors / Ligand-gated ion channel
2. Binding of neurotransmitter to receptors
3. Causes conformational change that open the opening of ligand-gated ion channel
4. allowing ions to cross the membrane into synaptic knob

OR

1. Acetylcholinesterase
2. Breaks down acetylcholine
3. into choline and acetate
4. Choline is reabsorbed into the synaptic knob to form acetylcholine again,

The binding of GABA to receptors on post-synaptic membrane causes negatively charged chloride ions to enter post-synaptic neurone through ligand-gated **Cl⁻ channels**.

(c) **Explain** how this will **inhibit production** of **nerve impulses** in the **post-synaptic neurone**. [2]

1. Inside of the cell body of post-synaptic neurone, become more negatively charged
2. becomes hyperpolarized
- Reduces effect of sodium ions entering
3. Does not reach threshold potential.
4. Action potential is not produced at the axon hillock.

Epilepsy may result when there is **increased neuronal activity** in the brain. One form of epilepsy has been linked to an abnormality in GABA receptors.

(d) **Suggest** how **abnormality of GABA** receptors results in epilepsy. [3]

1. Receptors that have different tertiary / 3D structure;
 2. Shape of binding site is not complementary to GABA;
 3. GABA cannot bind to GABA receptors;
 4. ligand-gated Cl⁻ channels do not open
 5. chloride ions do not enter
 6. Inhibition of neuronal activity does not occur
- Therefore there is increased neuronal activity in the brain.

[Total: 9]

SECTION B

Answer one question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 9 (a) Explain how the structure of the cell surface membrane facilitates bulk transport. [9]

1. The basic structure of the cell surface membrane is a phospholipid bilayer.
2. The hydrophilic phosphate heads of the phospholipids are in contact with the aqueous environment on either side of the membrane.
3. The hydrocarbon tails are shielded away from the aqueous environment and create a hydrophobic interior.
4. The fluidity of the membrane allows for bulk transport to occur. This is increased by
 5. more unsaturated the fatty acid tails which form kinks in the fatty acid tails,
 6. and presence of cholesterol
 7. that disturb the close packing of phospholipids.
8. Bulk transport mechanisms involve packaging in vesicles
9. and this requires energy/ a form of active transport.
10. Endocytosis is the process whereby the cells take in biological molecules and particulate matter (Accept: reference to Phagocytosis, pinocytosis)
 11. by forming new vesicles from the plasma membrane.
12. The plasma membrane extends outwards, forming extensions around the particles.
13. The plasma membrane can also invaginate (sinks inwards) to form a flask-like depression around the particles to engulf the particles.
14. Exocytosis is the process whereby the cells release/ secrete certain biological molecules
 15. by the fusion of vesicles with the plasma membrane.
16. When the vesicle membrane comes into contact with the plasma membrane, specific proteins rearrange the lipid molecules of the two bilayers so that the two membranes fuse and the contents of the vesicles are released out of the cell./ The vesicle membrane now becomes part of the plasma membrane.
17. Membrane proteins / Proteins embedded in the cell membrane function as receptors, where specific ligands can bind.

18. This allows for receptor-mediated endocytosis, which enables bulk quantities of specific substances to be taken in, to take place.

(b) Describe the general structures and functions of animal viruses.

[5]

Genetic material

1. The nucleic acid of a virus can be made up of DNA or RNA; Positive strand (+) RNA: the sequence of the viral genome is used directly for translation to synthesize viral proteins. Negative strand (-) RNA: the sequence of the viral genome is used as a template to be transcribed to form the viral mRNA before translation can occur.
2. Genetic material can be single-stranded or double-stranded;
3. The genome can be linear or circular;
4. Exist as a single copy or multiple copies in the virus
5. The nucleic acid contains viral genes that are responsible for coding viral proteins and viral enzymes necessary for replication / reproduction of viruses.

Shape of capsid

6. The capsid is the hollow protein coat that surrounds the viral genome and is composed of protein subunits called capsomeres.
7. The function of the capsid is to protect the viral genome,
8. And carry viral enzymes for replication purposes.
9. The shape of the capsid of a virus can be broadly classified as Icosahedral or Helical

Presence of outer envelope (only in certain viruses)

10. The viral envelope is derived from the plasma membrane of the host cell, which contains host cell phospholipids and membrane proteins.
11. It also has viral proteins and glycoproteins.
12. The function of the viral glycoproteins (spikes) that project from the envelope is to enable the viruses to infect their host cells by specifically identifying their host cells,
13. And binding to the host cell membrane (shape of glycoprotein is complementary to the shape of receptor protein on cell surface membrane of host cell),
14. and facilitating genome uptake into the host cells.

(c) Outline the process where bacteria undergo genetic recombination involving a named temperate phage.

[6]

1. Specialised transduction is associated with temperate phages eg. lambda (λ) phage
2. that integrate as a prophage into the bacterial donor chromosome,
3. during the lysogenic cycle.

4. When the phage DNA is excised from the donor bacterial chromosome, it sometimes
5. takes with it a small region of donor bacterial DNA that was adjacent to the phage DNA.
6. Thus, specialised transduction transfers only specific set of bacterial genes.
7. The phage DNA and the donor bacterial DNA is packaged into the viral capsid to form a specialised transducing particle.
8. A specialised transducing particle carries a hybrid DNA containing both bacterial and viral genetic material.
9. When the phage carrying the donor bacterial DNA infects a recipient bacterial cell,
10. the donor bacterial genes are injected along with the phage's DNA, thus transferring bacterial genes from donor to recipient cell.
11. The bacterial genes carried by the transducing particle recombine and integrate with the genes of the recipient bacterial cell.

10 (a) Describe the reproductive life cycle of HIV. [9]

Attachment/ adsorption and entry to host cell

1. The glycoprotein **gp120** on the surface of the enveloped HIV particle **specifically binds to cell surface protein CD4 (CD4 receptor)**
2. (host cell) **on the surface of CD4 T helper cells / macrophages.**
3. Once HIV binds to a host cell, the viral envelope fuses with the plasma membrane of the host cell, facilitated by gp 41.
4. In the cytoplasm, the viral capsid breaks down and releases its viral enzymes and RNA genome.

Replication of viral genome

(student should mention single-stranded (+) RNA molecules at least once in the answer)

These steps
take place in
the
cytoplasm

5. The two copies of single-stranded (+) RNA molecules are used as templates to synthesise two single-stranded complementary DNA (cDNA),
6. This process of transcribing RNA into DNA is called **reverse transcription**, and is carried out by the viral enzyme (HIV) reverse transcriptase.
7. Each cDNA strand is as a template to form linear double-stranded DNA (dsDNA) via complementary base pairing.
8. The newly formed double-stranded linear DNA then migrates to the nucleus,
9. where viral enzyme **Integrase** integrates the viral DNA into the host cell chromosome.
10. The provirus is replicated along with the host cell chromosome when the cell divides.
11. When these host cells get activated, proviral DNA gets transcribed into single-stranded (+) RNA molecules within the nucleus of the host cell.
12. These RNA molecules are then transported out of the nucleus into the cytoplasm of the host cell.
13. The (+) RNA strands are the viral genome for the new viral particles.

Synthesis of viral proteins

14. The complementary (+) RNA strands are used for the synthesis of viral proteins (such as capsid proteins, reverse transcriptase and glycoproteins gp120 and gp41)
The proteins are translated in the form of an inactive polypeptide chain
15. **HIV Protease** catalyses the hydrolysis of the polyprotein / polypeptide chain into individual functional proteins that are required for viral assembly and activity

Self-assembly of viral components (maturation) to new virus particles

16. Once formed, encapsidation of the viral genome occurs – capsid proteins assemble around the viral (+) single-stranded RNA molecules to form new nucleocapsids / new HIV particles.

Release/ exit of new virus particles from the host cell

17. Once formed, the viral glycoproteins gp120 and 41 are transported and incorporated into the plasma membrane of the host cell.
18. The newly assembled virus particles then bud off the host cell, acquiring the envelope with gp41 and gp120 embedded in the host cell membrane.

- (b) Describe the roles of membrane-bound proteins in a eukaryotic cell.
[5]

	Membrane-bound Protein	Role(s)
1.	Transport proteins:	(one mark for each type of transport protein) a. carrier proteins e.g GLUT (Glucose Transporter) transports only glucose and rejects fructose, a structural isomer b. channel proteins e.g aquaporin or hydrophilic channel on stalked particle for proton pass through c. ion pumps for active transport e.g. $\text{Na}^+\text{-K}^+$ pump, H^+ pump
2.	Glycoproteins	(any one role) a. cell-cell recognition: <u>is the ability of a cell to determine if other cells it encounters are alike or different from itself.</u> b. receptors, e.g. receptor mediated endocytosis, GPCR may have a <u>binding site</u> that binds to a chemical messenger (eg a hormone). This allows an <u>external signal</u> to trigger or initiate reactions within the cell. c. cell adhesion: Membrane proteins of adjacent cells may be attached together in various kinds <u>of intercellular junctions.</u> d. (Membrane protein) attachment to cytoskeleton, important in <u>maintaining cell shape</u>

3.	Enzymes attached to membranes	Catalyse a series of coordinated reactions in a sequential manner Adenyl cyclase that catalyse the conversion of ATP to cAMP.
4.	Electron carriers along ETC	Electrons are <u>transferred from electron donors to electron acceptors</u> (Accept: NADH or FADH ₂ to O ₂ ; PSII to NADP ⁺ ; PSI to PSI) by a <u>series of redox reactions</u> / <u>undergo redox reactions, chemiosmosis-ATP synthesis.</u>
5.	ATP synthase/ stalked particle	<u>Catalyze the formation of ATP from ADP and Pi in the matrix of mitochondria/ stroma of chloroplast.</u>
6.	NADP ⁺ reductase	<u>Reduce NADP⁺ to NADPH for carbon reduction in Calvin Cycle</u>

(c) Explain the role of membranes in the chloroplast. [6]

1. Chloroplasts have a double-membrane envelope
2. The envelope regulates the movement ions and metabolites into and out of the chloroplast with the presence of transport proteins embedded in the membrane.
3. The envelope also allows for compartmentalization of the reactions within the chloroplast, separated from the cytosol.
4. Chloroplasts also have internal membrane structures called thylakoids
5. Photosystems are found in the thylakoid membranes
6. These complexes enable light to be harvested for photophosphorylation/ light dependent reactions to occur at the membrane
7. Electron carriers, which make up the electron transport chain, are also found on the thylakoid membrane
8. For the transfer of electrons during photophosphorylation resulting in the reduction of NADP
9. The thylakoid membrane also encloses the thylakoid lumen / space, where a steep H⁺ gradient is maintained
10. to enable chemiosmosis to occur/ H⁺ diffuse through the stalked particles / ATP synthase embedded in the thylakoid membrane
11. resulting in ATP being formed from ADP and Pi

[Total: 20]

[Turn over

